REPUBLIQUE DU CAMEROUN
Paix – Travail – Patrie

UNIVERSITE DE YAOUNDE I

ECOLE NATIONALE SUPERIEURE POLYTECHNIQUE DE YAOUNDE (ENSPY)

REPUBLIC OF CAMEROON
Peace – Work – Fatherland

UNIVERSITY OF YAOUNDE I

NATIONAL ADVANCED SCHOOL OF ENGINEERING OF YAOUNDE

Rapport de Configuration – Cas LiZbiZ

Rapport académique

RÉALISÉ PAR :

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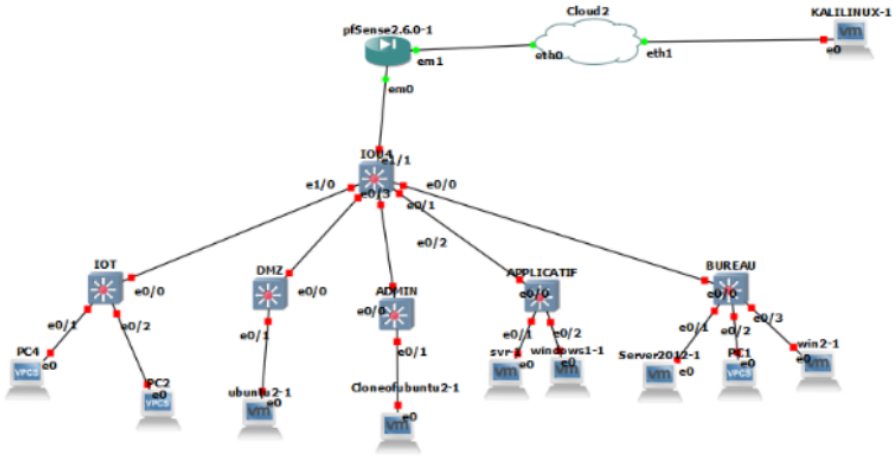
Informations de Mission

Mission : Sécuriser l'entreprise, auditer les pratiques et gérer les crises.
À l'attention de : COMEX
Date : Mars 2026
Classification : Confidentiel

1. Configuration proprement dite

Configuration VTP, VLAN, Trunk, Access et pare-feu.



1.1 Configurations de base

```
ernet1/0 (not full duplex), with IOT Ethernet0/1 (full duplex).
DIS#sh int status

Port      Name               Status       Vlan       Duplex  Speed Type
Et0/0                        connected    trunk        auto   auto unknown
Et0/1                        connected    trunk        auto   auto unknown
Et0/2                        connected    trunk        auto   auto unknown
Et0/3                        connected    99           auto   auto unknown
Et1/0                        connected    trunk        auto   auto unknown
Et1/1                        connected    99           auto   auto unknown
Et1/2                        connected    1            auto   auto unknown
Et1/3                        connected    1            auto   auto unknown
Et2/0                        connected    1            auto   auto unknown
Et2/1                        connected    1            auto   auto unknown
Et2/2                        connected    1            auto   auto unknown
Et2/3                        connected    1            auto   auto unknown
Et3/0                        connected    1            auto   auto unknown
Et3/1                        connected    1            auto   auto unknown
Et3/2                        connected    1            auto   auto unknown
Et3/3                        connected    1            auto   auto unknown
DIS#
*Mar 21 00:54:55.821: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethe
rnet0/0 (not full duplex), with DMZ Ethernet0/0 (full duplex).
DIS#
```

```
2.6.0-RELEASE][root@pfSense.home.arpa]/root: ifconfig em1
m1: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 mtu 1500
        options=81209b<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,VLAN_HWCSUM,WOL_MAG
C,VLAN_HWFILTER>
        ether 0c:55:4e:17:00:01
        inet6 fe80::e55:4eff:fe17:1%em1 prefixlen 64 scopeid 0x2
        inet 192.168.99.1 netmask 0xffffff00 broadcast 192.168.99.255
        media: Ethernet autoselect (1000baseT <full-duplex>)
        status: active
        nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
```

```
DIS#show mac address-table vlan 99
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
  99    0c55.4e17.0001    DYNAMIC     Et0/3
Total Mac Addresses for this criterion: 1
DIS#
*Mar 21 01:18:07.887: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethe
rnet0/2 (not full duplex), with BUREAU Ethernet0/0 (full duplex).
```

```
              Aging Time  300 sec

 Interface           Role Sts Cost      Prio.Nbr Type
 ------------------- ---- --- --------- -------- -------------------------------
 Et0/0               Root FWD 100       128.1    Shr
 Et0/1               Desg FWD 100       128.2    Shr
 Et0/2               Desg FWD 100       128.3    Shr
 Et0/3               Desg FWD 100       128.4    Shr
 Et1/0               Desg FWD 100       128.5    Shr
 Et1/1               Desg FWD 100       128.6    Shr Edge

  --More--
 *Mar 21 01:21:13.770: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethe
 rnet1/0 (not full duplex), with IOT Ethernet0/1 (full duplex).
 *Mar 21 01:21:14.026: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethe
```

```
IOU4#vtp mode server
         ^
% Invalid input detected at '^' marker.

IOU4#vtp domain AUDIT-CRISIS
         ^
% Invalid input detected at '^' marker.

IOU4#vtp password cisco123
Setting device VTP password to cisco123
IOU4#vtp version 2
IOU4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
IOU4(config)#vtp domain LIZBIZ
Changing VTP domain name from NULL to LIZBIZ
IOU4(config)#
```

```
IOU4(config)#vlan 10
IOU4(config-vlan)# name DMZ
IOU4(config-vlan)#vlan 20
IOU4(config-vlan)# name Applicatif
IOU4(config-vlan)#vlan 30
IOU4(config-vlan)# name Bureau
IOU4(config-vlan)#vlan 40
IOU4(config-vlan)# name IoT
IOU4(config-vlan)#vlan 99
IOU4(config-vlan)# name Admin
IOU4(config-vlan)#exit
IOU4(config)#
```

```
IOU4(config-if)#interface e0/2
IOU4(config-if)# switchport trunk encapsulation dot1q
IOU4(config-if)# switchport mode trunk
IOU4(config-if)# switchport trunk allowed vlan 10,20,30,40,99
IOU4(config-if)# no shutdown
IOU4(config-if)#
```

```
*Mar 22 19:45:18.049: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, ch
anged state to up
IOU4(config-if)#interface e0/0
IOU4(config-if)# switchport trunk encapsulation dot1q
IOU4(config-if)# switchport mode trunk
IOU4(config-if)# switchport trunk allowed vlan 10,20,30,40,99
IOU4(config-if)# no shutdown
IOU4(config-if)#
*Mar 22 19:45:26.400: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, ch
anged state to down
IOU4(config-if)#
*Mar 22 19:45:29.409: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, ch
anged state to up
```

```
OU4(config-if)#interface vlan 99
OU4(config-if)# ip address 192.168.99.2 255.255.255.0
OU4(config-if)# no shutdown
OU4(config-if)#
Mar 22 19:45:55.486: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed
state to down
OU4(config-if)#
Mar 22 19:45:57.786: %LINK-3-UPDOWN: Interface Vlan99, changed state to up
Mar 22 19:45:58.792: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed
state to up
OU4(config-if)#ip default-gateway 192.168.99.1
OU4(config)#end
OU4#w
Mar 22 19:46:13.666: %SYS-5-CONFIG_I: Configured from console by console
OU4#wr memory
arning: Attempting to overwrite an NVRAM configuration previously written
y a different version of the system image.
verwrite the previous NVRAM configuration?[confirm]
uilding configuration...
ompressed configuration from 2096 bytes to 1158 bytes[OK]
```

```
OT#conf t
nter configuration commands, one per line.  End with CNTL/Z.
OT(config)#hostname IOT
OT(config)#vtp mode client
etting device to VTP Client mode for VLANS.
OT(config)#vtp domain LIZBIZ
hanging VTP domain name from NULL to LIZBIZ
OT(config)#vtp password cisco123
etting device VTP password to cisco123
OT(config)#vtp version 2
```

```
IOT(config-if)# no shutdown
IOT(config-if)#interface e0/2
IOT(config-if)# switchport mode access
IOT(config-if)# switchport access vlan 40
IOT(config-if)#no shut
IOT(config-if)#
```

```
DMZ(config)#
DMZ(config)#interface e0/0
DMZ(config-if)# switchport trunk encapsulation dot1q
DMZ(config-if)# switchport mode trunk
DMZ(config-if)# switchport trunk allowed vlan 10,20,30,40,99
DMZ(config-if)# no shutdown
DMZ(config-if)#interface e0/1
DMZ(config-if)# switchport mode access
DMZ(config-if)# switchport access vlan 10
DMZ(config-if)#no shut
DMZ(config-if)#interface vlan 99
DMZ(config-if)# ip address 192.168.99.4 255.255.255.0
DMZ(config-if)# no shutdown
DMZ(config-if)#
*Mar 22 19:54:17.429: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, ch
anged state to down
DMZ(config-if)#
*Mar 22 19:54:19.538: %LINK-3-UPDOWN: Interface Vlan99, changed state to up
*Mar 22 19:54:20.542: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, ch
anged state to up
DMZ(config-if)#ip default-gateway 192.168.99.1
DMZ(config)#
```

1.2 Configuration du pare-feu

Adressage SVI, NAT et règles de sécurité

```
APPLICATIF(config)#vtp mode client
Setting device to VTP Client mode for VLANS.
APPLICATIF(config)#vtp domain cisco123
Changing VTP domain name from NULL to cisco123
APPLICATIF(config)#vtp domain LIZBIZ
Changing VTP domain name from cisco123 to LIZBIZ
APPLICATIF(config)#vtp password cisco123
Setting device VTP password to cisco123
APPLICATIF(config)#vtp version 2
```

```
APPLICATIF(config)#interface e0/0
APPLICATIF(config-if)# switchport trunk encapsulation dot1q
APPLICATIF(config-if)# switchport mode trunk
APPLICATIF(config-if)# switchport trunk allowed vlan 10,20,30,40,99
APPLICATIF(config-if)# no shutdown
APPLICATIF(config-if)#interface e0/1
APPLICATIF(config-if)# switchport mode access
APPLICATIF(config-if)# switchport access vlan 20
APPLICATIF(config-if)#no shut
APPLICATIF(config-if)#interface e0/2
APPLICATIF(config-if)# switchport mode access
APPLICATIF(config-if)# switchport access vlan 20
APPLICATIF(config-if)#no shutdown
APPLICATIF(config-if)#
```

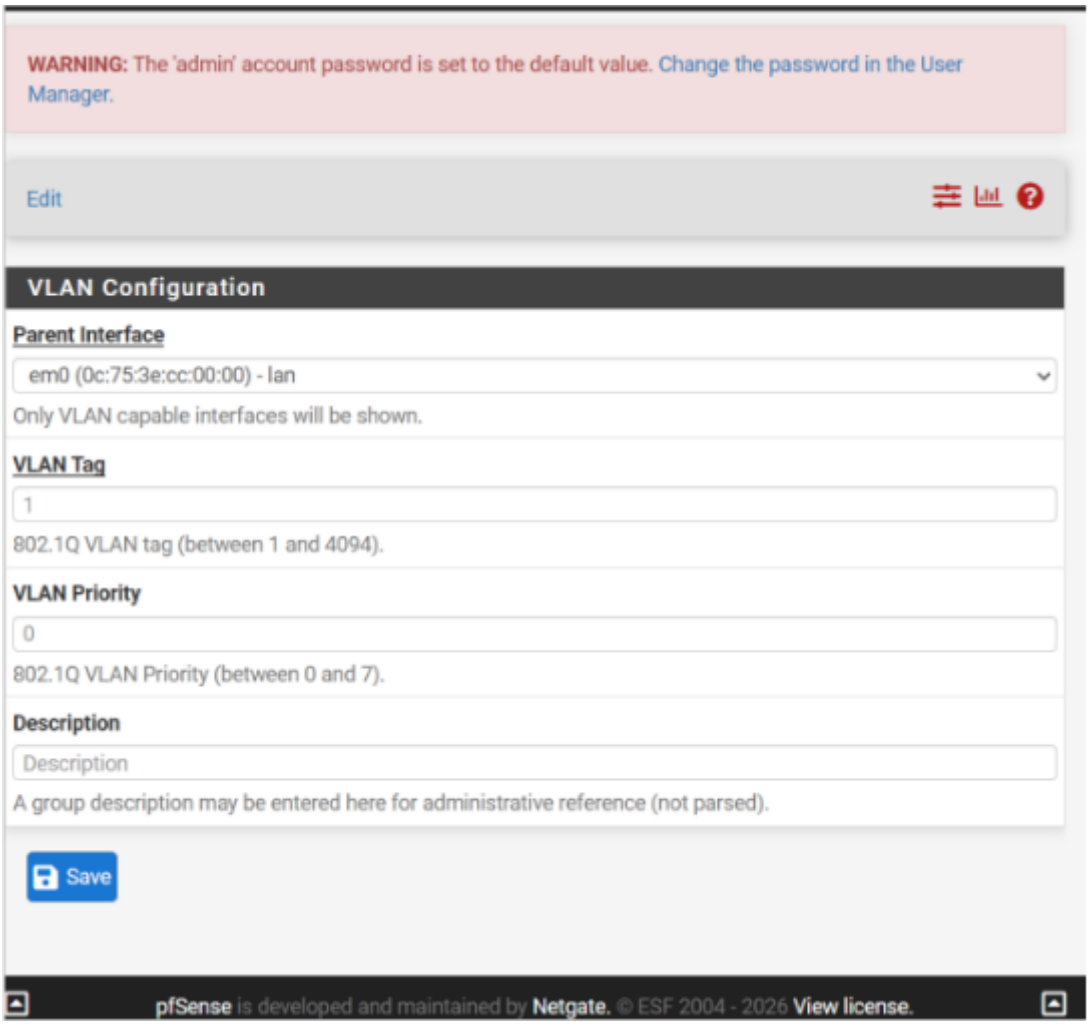
```
nter the number of the interface you wish to configure: 2

nter the new LAN IPv4 address.  Press <ENTER> for none:
 192.168.99.1

ubnet masks are entered as bit counts (as in CIDR notation) in pfSense.
.g. 255.255.255.0 = 24
    255.255.0.0   = 16
    255.0.0.0     = 8

nter the new LAN IPv4 subnet bit count (1 to 32):
 24

or a WAN, enter the new LAN IPv4 upstream gateway address.
or a LAN, press <ENTER> for none:
```

```
IOT#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/3, Et1/0, Et1/1, Et1/2
                                                Et1/3, Et2/0, Et2/1, Et2/2
                                                Et2/3, Et3/0, Et3/1, Et3/2
                                                Et3/3
10   DMZ                              active
20   Applicatif                       active
30   Bureau                           active
40   IoT                              active    Et0/1, Et0/2
99   Admin                            active
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup
IOT#
```

```
IOU4#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
BUREAU           Eth 0/0           173             R S I  Linux Uni Eth 0/0
APPLICATIF       Eth 0/1           167             R S I  Linux Uni Eth 0/0
ADMIN            Eth 0/2           157             R S I  Linux Uni Eth 0/0
IOT              Eth 1/0           144             R S I  Linux Uni Eth 0/0
DMZ              Eth 0/3           136             R S I  Linux Uni Eth 0/0

Total cdp entries displayed : 5
IOU4#
```

```
Port      Name               Status       Vlan       Duplex  Speed Type
Et0/0                        connected    trunk        auto   auto unknown
Et0/1                        connected    trunk        auto   auto unknown
Et0/2                        connected    trunk        auto   auto unknown
Et0/3                        connected    trunk        auto   auto unknown
Et1/0                        connected    trunk        auto   auto unknown
Et1/1                        connected    trunk        auto   auto unknown
Et1/2                        connected    1            auto   auto unknown
Et1/3                        connected    1            auto   auto unknown
Et2/0                        connected    1            auto   auto unknown
Et2/1                        connected    1            auto   auto unknown
Et2/2                        connected    1            auto   auto unknown
```

```
er an option: 8

6.0-RELEASE][root@pfSense.home.arpa]/root: ifconfig em1 inet 10.0.0.2 netmask
5.255.255.252 up
6.0-RELEASE][root@pfSense.home.arpa]/root: if config em0 inet 192.168.99.1 ne
sk 255.255.255.0 up
 Expression Syntax.
6.0-RELEASE][root@pfSense.home.arpa]/root: ifconfig em0 inet 192.168.99.1 net
k 255.255.255.0 up
6.0-RELEASE][root@pfSense.home.arpa]/root:
```


WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Edit
VLAN Configuration
Parent Interface
em0 (0c:75:3e:cc:00:00) - lan
Only VLAN capable interfaces will be shown.
VLAN Tag
1
802.1Q VLAN tag (between 1 and 4094).
VLAN Priority
0
802.1Q VLAN Priority (between 0 and 7).
Description
Description
A group description may be entered here for administrative reference (not parsed).
Save
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Floating WAN LAN DMZ APPLICATIF BUREAU IOT ADMIN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Descri
✓	0 /0 B	IPv4 *	BUREAU net	*	*	*	*	none		burea
✓	0 /0 B	IPv4 TCP	BUREAU net	*	APPLICATIF net	PORTS_ APPLI	*	none		burea
✗	0 /0 B	IPv4 *	BUREAU net	*	IOT net	*	*	none		burea
✗	0 /0 B	IPv4 *	BUREAU net	*	IOT net	*	*	none		burea
✗	0 /0 B	IPv4 *	BUREAU net	*	ADMIN net	*	*	none		burea

Floating WAN LAN DMZ APPLICATIF BUREAU IOT ADMIN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Des
✗	0 /0 B	IPv4 *	IOT net	*	APPLICATIF net	*	*	none		iot
✗	0 /0 B	IPv4 *	DMZ net	*	APPLICATIF net	*	*	none		dn
✗	0 /0 B	IPv4 *	WAN net	*	*	*	*	none		pa
✓	0 /0 B	IPv4 *	ADMIN net	*	APPLICATIF net	*	*	none		ad
✓	0 /0 B	IPv4 *	APPLICATIF net	*	APPLICATIF net	*	*	none		AF
✓	0 /0 B	IPv4 TCP	BUREAU net	*	APPLICATIF net	PORTS_ APPLI	*	none		BU

1.3 Test de validation

```
PC1>
PC1> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=64 time=57.988 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=64 time=3.317 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=64 time=11.059 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=64 time=10.271 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=64 time=11.087 ms

PC1>
```

```
PC1> ping 192.168.99.1

192.168.99.1 icmp_seq=1 timeout
192.168.99.1 icmp_seq=2 timeout
192.168.99.1 icmp_seq=3 timeout
192.168.99.1 icmp_seq=4 timeout
192.168.99.1 icmp_seq=5 timeout

PC1> ping 192.168.40.1

192.168.40.1 icmp_seq=1 timeout
192.168.40.1 icmp_seq=2 timeout
192.168.40.1 icmp_seq=3 timeout
192.168.40.1 icmp_seq=4 timeout
192.168.40.1 icmp_seq=5 timeout

PC1>
```

```
192.168.99.1 icmp_seq=1 timeout
192.168.99.1 icmp_seq=2 timeout
192.168.99.1 icmp_seq=3 timeout
192.168.99.1 icmp_seq=4 timeout
192.168.99.1 icmp_seq=5 timeout

PC1> ping 192.168.40.1

192.168.40.1 icmp_seq=1 timeout
192.168.40.1 icmp_seq=2 timeout
192.168.40.1 icmp_seq=3 timeout
192.168.40.1 icmp_seq=4 timeout
192.168.40.1 icmp_seq=5 timeout

PC1> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=64 time=21.353 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=64 time=12.655 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=64 time=15.223 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=64 time=51.877 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=64 time=15.136 ms
```

```
192.168.205.2       0c:75:3e:cc:00:01  UHS          em1
192.168.205.152     link#2             UHS          lo0

Internet6:
Destination                        Gateway                        Flags      Netif
Expire
::1                                link#8                         UH           lo0
fe80::%em0/64                      link#1                         U           em0
fe80::1:1%em0                      link#1                         UHS          lo0
fe80::e75:3eff:fecc:0%em0          link#1                         UHS          lo0
fe80::%em1/64                      link#2                         U           em1
fe80::e75:3eff:fecc:1%em1          link#2                         UHS          lo0
fe80::%lo0/64                      link#8                         U            lo0
fe80::1%lo0                        link#8                         UHS          lo0
fe80::%em0.10/64                   link#11                        U        em0.10
fe80::e75:3eff:fecc:0%em0.10       link#11                        UHS          lo0
fe80::%em0.20/64                   link#12                        U        em0.20
fe80::e75:3eff:fecc:0%em0.20       link#12                        UHS          lo0
fe80::%em0.30/64                   link#13                        U        em0.30
fe80::e75:3eff:fecc:0%em0.30       link#13                        UHS          lo0
fe80::%em0.40/64                   link#14                        U        em0.40
fe80::e75:3eff:fecc:0%em0.40       link#14                        UHS          lo0
fe80::%em0.99/64                   link#15                        U        em0.99
fe80::e75:3eff:fecc:0%em0.99       link#15                        UHS          lo0
[2.6.0-RELEASE][root@pfSense.home.arpa]/root:
```

En gros, voilà ce qui en ressort de nos analyses.

2. Conclusion

L'entreprise LiZbiZ se trouve à un tournant stratégique de son développement. La croissance rapide qu'elle a connue n'a pas été accompagnée d'une structuration adéquate de sa gouvernance de la sécurité, laissant l'entreprise exposée à des risques majeurs.